

Introduction to Electronic Design Automation

Spring 2026

National Taiwan University

Problem Set 2

Problems 1-5 due on 2026/04/21 23:59; Problems 6-8 due on 2026/04/28 23:59.

1 [Cofactor]

[10%] Given two Boolean functions f and g and literals u and v , prove or disprove the following equalities.

- (a) (3%) $(f_u)_v = (f_v)_u$
- (b) (3%) $(\neg f)_v = \neg(f_v)$
- (c) (4%) $(f \oplus g)_v = (f_v) \oplus (g_v)$

2 [Quantified Boolean Formula]

[15%] Given two arbitrary Boolean functions f and g , please justify which implication directions ($\Leftarrow$ and/or $\Rightarrow$) of the following arrows $\Leftrightarrow$ hold generally, and show counterexamples to those that do not hold.

- (a) (3%)
$$\neg(\forall x, \exists y. f(x, y, z)) \Leftrightarrow \exists x, \forall y. \neg f(x, y, z)$$
- (b) (3%)
$$\exists x, \forall y. f(x, y, z) \Leftrightarrow \forall x, \exists y. f(x, y, z)$$
- (c) (3%)
$$\exists x, \forall y. f(x, y, z) \Leftrightarrow \forall y, \exists x. f(x, y, z)$$
- (d) (3%)
$$\forall x. (f(x, y) \wedge g(x, z)) \Leftrightarrow (\forall x. f(x, y)) \wedge (\forall x. g(x, z))$$
- (e) (3%)
$$\forall x. (f(x, y) \vee g(x, z)) \Leftrightarrow (\forall x. f(x, y)) \vee (\forall x. g(x, z))$$

3 [Boolean Function Representation]

[15%] Consider the majority function $f(x_1, \dots, x_5) = (x_1 + x_2 + \dots + x_5 \geq 3)$.

- (a) (3%) Represent f in terms of a minimum sum-of-products (SOP) expression.
- (b) (3%) Represent f in terms of a minimum product-of-sums (POS) expression.
- (c) (3%) Represent f in terms of an ROBDD.
- (d) (3%) Represent f in terms of an AIG with as few nodes (i.e., 2-input AND gates) as possible.
- (e) (3%) Represent f in terms of a network of 3-input majority gates. (A 3-input majority gate $M(a, b, c) = 1$ if and only if at least two among the inputs a, b, c are 1).

4 [Application of Quantified Boolean Formula]

[12%] Consider a simplified version of the Nim game played with 3 piles of stones. Initially, the piles contain the following number of stones: Pile 1: 2 stones, Pile 2: 2 stones, and Pile 3: 1 stone. Two players, Player P_1 and Player P_2 , take turns removing stones from the piles. On each turn, a player must remove at least one stone from one pile, but they can remove any number of stones from that pile. The player who removes the last stone from the piles wins the game. Player P_1 goes first, followed by Player P_2 , Player P_1 again, and so on, until no stone left.

Your task is to determine whether Player P_1 has a winning strategy, that is, a way to choose his/her moves so that he/she can always win, regardless of what moves Player P_2 makes. To do this, use a Quantified Boolean Formula (QBF) to model the game and analyze whether Player P_1 has a strategy that guarantees victory.

- (a) (6%) Write a QBF formula to encode the game, and explain the intuition behind the formula.
 - Define Boolean variables to represent Player P_1 's and P_2 's choices in each move.
 - Encode the game's rules as constraints on these variables, including the alternation of turns and the condition for winning.
 - Explain the reasoning behind the variables and constraints, and how they model the sequence of moves leading to Player P_1 's or P_2 's potential victory.
- (b) (6%) Determine the truth or falsity of the formula, and the corresponding winning strategy of the winning player.
 - Use a QBF solver or manual reasoning to evaluate the truth of the formula.
 - If the formula is true, explain the winning strategy for Player P_1 and describe how Player P_1 can always win regardless of Player P_2 's responses.
 - If the formula is false, explain why Player P_1 does not have a guaranteed winning strategy, and describe the strategy Player P_2 can use to prevent Player P_1 from winning.

Hint: 1) Use quantifiers to model the alternating moves between Players P_1 and P_2 . 2) Define the moves as Boolean variables for the number of stones removed from each pile during each turn, and constrain the legal move sequences. 3) Model the game's termination condition (when there are no stones left) and the winning condition (removing the last stone).

5 [Binary Decision Diagram]

[12%]

- (a) (6%) Prove that ROBDD with respect to a fixed variable order is a canonical representation of Boolean functions. (Hint: Show by induction.)

- (b) (6%) Consider building the ROBDD for the Boolean function $f = \bigwedge_{i=1}^n (x_i \vee y_i)$. Find a variable order and build the corresponding ROBDD with $O(n)$ nodes. Find another variable order and build the corresponding ROBDD with $O(2^n)$ nodes.

6 [Two-level Logic Minimization]

[18%] Let

$$G = abce + abde + abc'e + a'bc'e + a'b'c'e' + a'c'd'e + bcd'e + cd'e'$$

be a cover for an incompletely-specified Boolean function (F, D, R) with

$$D = a'bd'e + abcd' + ace'$$

being a cover for the characteristic function of the don't-care set D. Apply Quine-McCluskey's algorithm to find a minimum cover G^* for the incompletely-specified Boolean function and answer the following questions.

- (3%) Derive the minterms of the offset R.
- (3%) Generate all prime implicants, and list the Boolean matrix.
- (3%) Which prime implicants are essential? Reduce the Boolean matrix based on the prime implicants.
- (3%) Reduce iteratively the Boolean matrix using row equality, row dominance, column dominance, and induced essential primes until no more reduction is possible. Show intermediate steps.
- (3%) Solve the cyclic core, if any, using the independent set heuristic algorithm if applicable.
- (3%) Derive the minimum cover G^* for the incompletely-specified Boolean function.

7 [Static Timing Analysis]

[10%]

- (5%) Given a combinational circuit, show that the circuit region(s) with slack values $\leq c$, for constant c greater than or equal to the minimum slack of the circuit, must consist of paths from primary inputs to primary outputs.
- (5%) Perform static timing analysis on the circuit of Figure 1. Assume the target clock cycle is 16ns. Compute the *arrival time*, *required time*, and *slack* of *every wire* (note that a multi-terminal net consists of a set of *two-terminal* wires connecting pins and Steiner points). Identify critical path(s) with slacks less than 0ns.

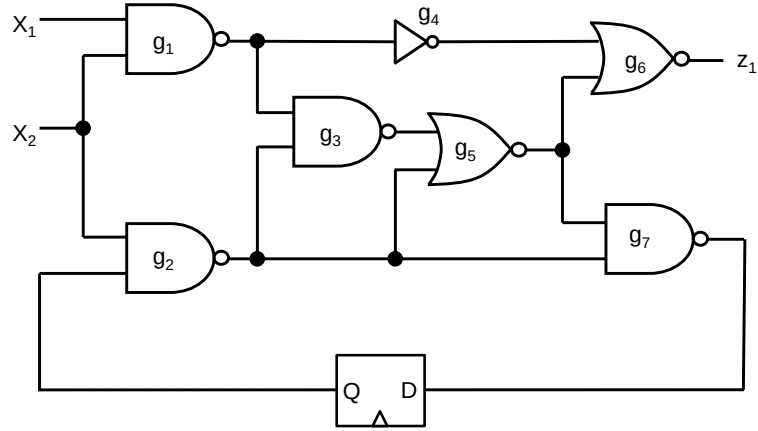


Fig. 1. A circuit under static timing analysis. Assume the arrival time of x_1 is 0ns, that of x_2 is 2ns, the propagation delay of the D-flip-flop is 1ns, the setup time of the flip-flop is 2ns, and the gate delays of inverter, NAND, and NOR are 1ns, 3ns, and 4ns, respectively.)

8 [Technology Mapping]

[8%] Consider the subject graph and pattern graphs in Figure 2 and Figure 3, respectively. Let the number in the parentheses of a pattern graph in Figure 3 be the area cost. Apply the DAGON mapping algorithm to find a cover of minimum area. Show all the optimal costs with their corresponding cell names at every node, and identify the final optimum cover.

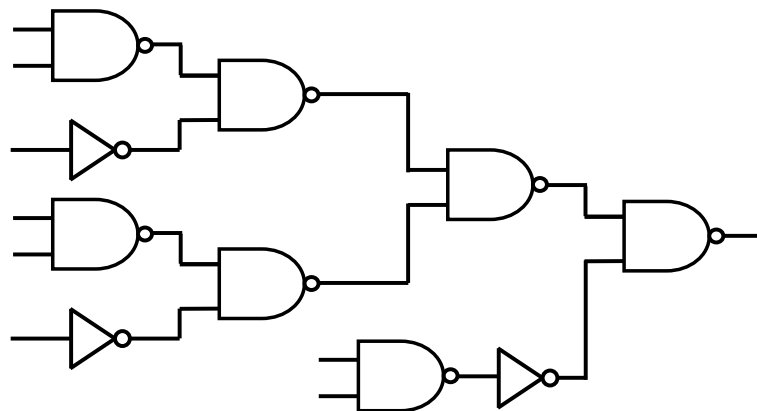


Fig. 2. Subject graph

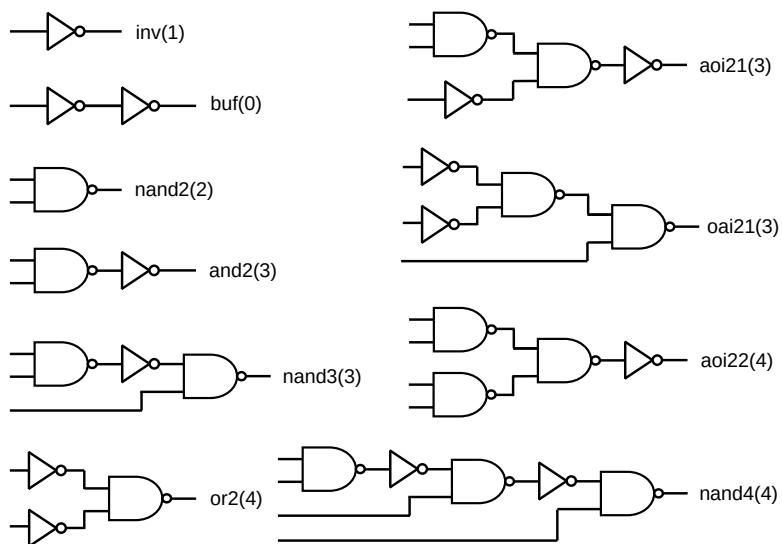


Fig. 3. Pattern graphs.